Program-8 : Create Kubernetes Pods using YAML Descriptors

Description :
This program demonstrates how to create and run a Kubernetes Pod using YAML descriptors. A Node.js application is deployed inside a Pod using a ConfigMap and accessed using port forwarding.

Step-1 : Start Minikube
```
minikube start --driver=docker
```

Step-2 : Create Project Directory
```
cd ~
mkdir program-8
cd program-8
```

Step-3 : Create Node.js Application File
```
nano server.js
```

```
const http = require("http");
const port = 3000;

const server = http.createServer((req, res) => {
  res.end("Hello from Node.js running inside Kubernetes!");
});

server.listen(port, () => {
  console.log(`Server running at port ${port}`);
});
```

Step-4 : Create ConfigMap YAML File
```
nano nodejs-configmap.yaml
```

```
apiVersion: v1
kind: ConfigMap
metadata:
  name: nodejs-app-config
data:
  server.js: |
    const http = require("http");
    const port = 3000;

    const server = http.createServer((req, res) => {
      res.end("Hello from Node.js running inside Kubernetes!");
    });
```

```
  server.listen(port, () => {
    console.log(`Server running at port ${port}`);
  });
```

Step-5 : Create Pod YAML File
nano nodejs-pod.yaml

```
apiVersion: v1
kind: Pod
metadata:
  name: program-8
spec:
  containers:
  - name: nodejs
    image: node:18
    command: ["node", "/usr/src/app/server.js"]
    ports:
    - containerPort: 3000
    volumeMounts:
    - name: app-code
      mountPath: /usr/src/app
  volumes:
  - name: app-code
    configMap:
      name: nodejs-app-config
```

Step-6 : Apply ConfigMap
kubectl apply -f nodejs-configmap.yaml

Step-7 : Apply Pod
kubectl apply -f nodejs-pod.yaml

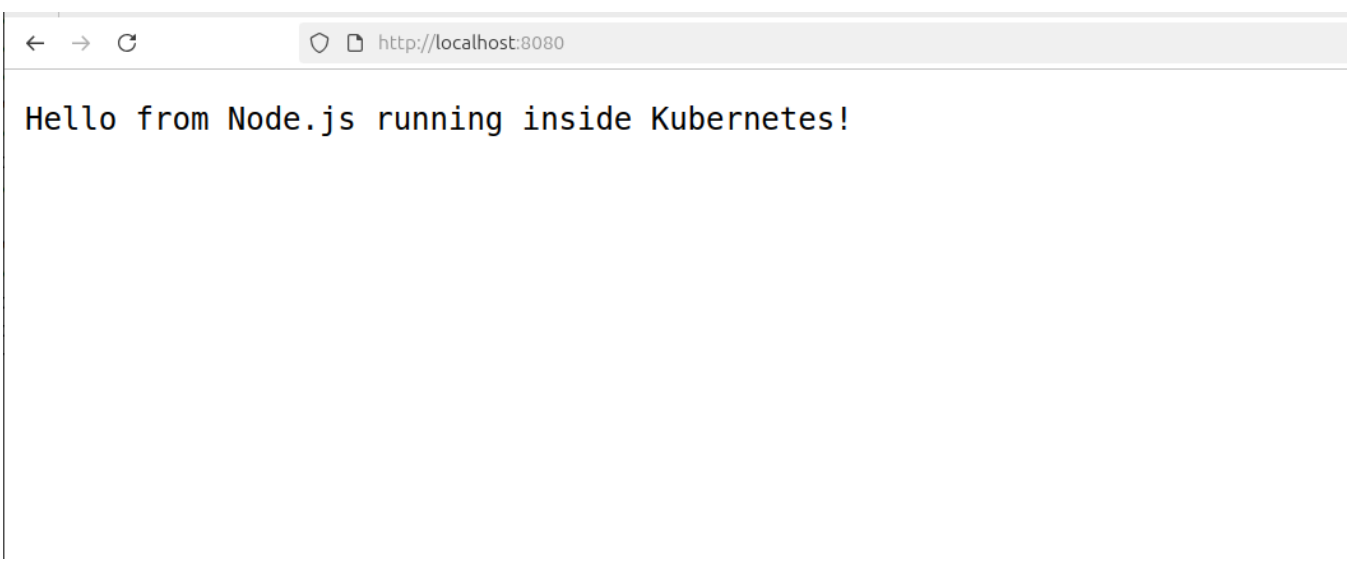
Step-8 : Check Pod Status
kubectl get pods

```
pavan-k@pavank1RV24MC072:~/program-8$ cd ~/program-8
pavan-k@pavank1RV24MC072:~/program-8$ nano server.js

pavan-k@pavank1RV24MC072:~/program-8$ nano nodejs-configmap.yaml
pavan-k@pavank1RV24MC072:~/program-8$ nano nodejs-pod.yaml
pavan-k@pavank1RV24MC072:~/program-8$ kubectl apply -f nodejs-configmap.yaml
configmap/nodejs-app-config created
pavan-k@pavank1RV24MC072:~/program-8$ kubectl apply -f nodejs-pod.yaml
pod/program-8 created
pavan-k@pavank1RV24MC072:~/program-8$ kubectl get pods
NAME        READY   STATUS    RESTARTS   AGE
program-8   1/1     Running   0          8s
pavan-k@pavank1RV24MC072:~/program-8$ kubectl port-forward pod/program-8 8080:3000
Forwarding from 127.0.0.1:8080 -> 3000
Forwarding from [::1]:8080 -> 3000
Handling connection for 8080
```

Step-9 : Access Application using Port Forwarding
kubectl port-forward pod/program-8 8080:3000

Step-10 : Verify Output
Open browser and visit:
http://localhost:8080



Output:
Hello from Node.js running inside Kubernetes!

Step-11 : Cleanup Resources

kubectl delete pod program-8
kubectl delete configmap nodejs-app-config

Result :
A Kubernetes Pod was successfully created using YAML descriptors. The Node.js application was deployed using a ConfigMap and accessed through port forwarding.